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Education

- Tsinghua University — PhD Candidate, Chemistry** 2023.09 – present
Research focus: chemical biology, targeted protein degradation (PROTAC), AI-guided protein design
- Sun Yat-sen University — M.S., Basic Medicine** 2020.09 – 2023.07
Research focus: cancer molecular mechanisms, protein post-translational modification, inflammation–cancer crosstalk
- Northwest A&F University — B.S., Animal Science** 2016.09 – 2020.07
Foundational training in animal experimentation, histology, physiology and molecular biology

Research Experience

PhD dissertation research

- Dual-target bioPROTAC design and endocrine-resistance mechanisms** 2023.09 – present
- Designed a tumor-specific, DNA-encoded bioPROTAC co-degrading ER α and PI3K p85 α to re-sensitize tamoxifen-resistant breast cancer cells to endocrine therapy;
 - Validated dose- and time-dependent dual-target degradation (Western blot) and demonstrated significant synergy with tamoxifen in resistant cells (CI < 0.4);
 - Characterized downstream apoptosis and cell-cycle-arrest phenotypes to establish a mechanistic link between degradation and re-sensitization.
- KRAS-targeting bioPROTAC and a degradation-escape mechanism** 2024.09 – present
- Engineered a KRAS-targeting bioPROTAC on a TRIM21-RBCC E3 scaffold and identified a resistance-associated driver mutation in the KRAS hypervariable region that causes degradation escape from this multimerization-dependent platform;
 - Traced the mechanism using 200-ns all-atom molecular dynamics simulations of the farnesylated, membrane-embedded protein (GROMACS/AMBER, CHARMM-GUI), revealing mutation-driven Switch I/II conformational opening;
 - Functionally confirmed the multimerization-dependence hypothesis using an orthogonal, multimerization-independent degrader as a mechanistic control;
 - Performed cross-validated (dual search-engine) proteomic and ubiquitomic profiling to characterize pathway-level features of the resistant background.

Independent computational research

- Reproduction of the CLAIRE de novo protein-binder design pipeline (Kortemme lab)** 2026.02 – present
- Independently reproduced the CLAIRE pipeline (Rosetta motif generation/matching, ProteinMPNN, AlphaFold2 self-consistency validation) end-to-end for three small-molecule targets — progesterone (the repository's built-in example), thalidomide, and 4-hydroxytamoxifen (4-OHT) — the latter two deliberately chosen to stress-test generalization across chemically distinct polar-handle configurations;
 - Built the full Rosetta/PyRosetta/ColabFold compute environment from scratch; diagnosed and fixed a hidden file-naming defect in the original codebase;
 - Produced AlphaFold2-self-consistent candidates for all three targets (best case: pLDDT 94.1, RMSD 0.62 Å) and published the full reproduction, including methodology and debugging narrative, publicly on GitHub.

Testing the generalizability of a multistate/dynamic protein-design method (Kortemme lab) on KRAS

2026.08 – present

- Extracted the exact computational recipe of Guo et al. (Science, 2025) — an AF2-based conformational-state reversion scan combined with position-tied ProteinMPNN — directly from the primary literature;
- Designed a three-layer validation framework to test whether this diagnostic generalizes to KRAS/Ras-superfamily GTPases: a calibration layer (T35S, a literature-established state-switching mutant), a clinical-noise layer (G12D/G12C), and an application layer (the hypervariable-region mutation identified in my own dissertation work);
- Rebuilt a nucleotide- and Mg²⁺-bound (GppNHp) KRAS4B membrane simulation system in AMBER after diagnosing a structural defect (a nucleotide-free system) in a prior model, including de novo force-field parameterization for the non-standard GppNHp ligand and the farnesylated cysteine membrane anchor.

Prior research experience

PGAM5 O-GlcNAcylation and tumor progression

2022.09 – 2023.06

- Characterized the PGAM5–OGT interaction and its role in protein modification, stability and tumor invasion; built a PGAM5-knockout model to analyze inflammation-related phenotypes and tumor adaptation.

National Natural Science Foundation of China (NSFC) project

2021.01 – 2023.07

- P311 regulation of Periostin translation in pancreatic cancer metastasis: molecular mechanism and functional study (Grant No. 32000543).

SARS-CoV-2 replicon system and nsp functional analysis

2020.09 – 2022.09

- Built nsp deletion and mutation models to systematically evaluate viral replication competence;
- Identified nsp13-C16446T as an enhancer of the nsp12 interaction with a significant effect on the replication mechanism.

Publications

1. Ouyang M, Feng Y, Chen H, Liu, Tan C, Tan Y. Recent advances in optically controlled PROTAC. *Bioengineering*. 2023;10(12):1368.
2. Ouyang M, Yu C, Deng X, Zhang Y, Zhang X, Duan F. O-GlcNAcylation and its role in cancer-associated inflammation. *Frontiers in Immunology*. 2022;13:861559.
3. Zhang C, Wang J, Chen H, Ouyang M, Li Y, Tan C, Jiang Y, Tan Y. A Therapeutic Pretargeting Strategy for Triple-Negative Breast Cancer Based on a Bioorthogonal Reaction and a Self-Assembled Peptide. *Advanced Functional Materials*. 2024;34(24):2313090.
4. Liang J, Deng X, Zhang Y, Fei T, Ouyang M, Yu C, Xiang Y, Jia D, Duan F. Bioinformatics insights into the role of GFPT1 in breast invasive carcinoma: implications for tumor prognosis, immune modulation, and therapeutic applications. *Frontiers in Genetics*. 2024;15:1482929.
5. Guan H, Zhong M, Ma K, Tang C, Wang X, Ouyang M, Qin R, Chen J, et al. The comprehensive role of high mobility group box 1 (HMGB1) protein in different tumors: a pan-cancer analysis. *Journal of Inflammation Research*. 2023:617–637.
6. Jin Y, Ouyang M, Yu T, Zhuang J, Wang W, Liu X, Duan F, Guo D, Peng X, et al. Genome-wide analysis of the indispensable role of non-structural proteins in the replication of SARS-CoV-2. *Frontiers in Microbiology*. 2022;13:907422.
7. Duan F, Ouyang M, Jia D, Ma K, Pan J. IDDF2021-ABS-0072 P311 promotes M2 macrophage polarization and infiltration in pancreatic ductal adenocarcinoma. *Gut*. 2021;70(Suppl 2):A27–A28.

Technical Skills

- Computational protein design: Rosetta/PyRosetta, ProteinMPNN, AlphaFold2/ColabFold.
- Molecular dynamics & modeling: GROMACS, AMBER/AmberTools, OpenMM, CHARMM-GUI, force-field parameterization (antechamber, AM1-BCC, GAFF2).
- Omics data analysis: whole-genome sequencing (WGS) variant calling, DIA mass-spectrometry quantification.
- Wet-lab techniques: mammalian cell culture, Western blot, flow cytometry (apoptosis/cell-cycle analysis), plasmid construction and transfection, siRNA knockdown, co-immunoprecipitation, chemical cross-linking, immunohistochemistry, subcellular fractionation, Sanger sequencing, animal-experiment design.
- Computing: Python scientific computing, Linux/cloud GPU environment setup and orchestration.